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BFIS-02

O(1) Constant time

Example 1



Example 2

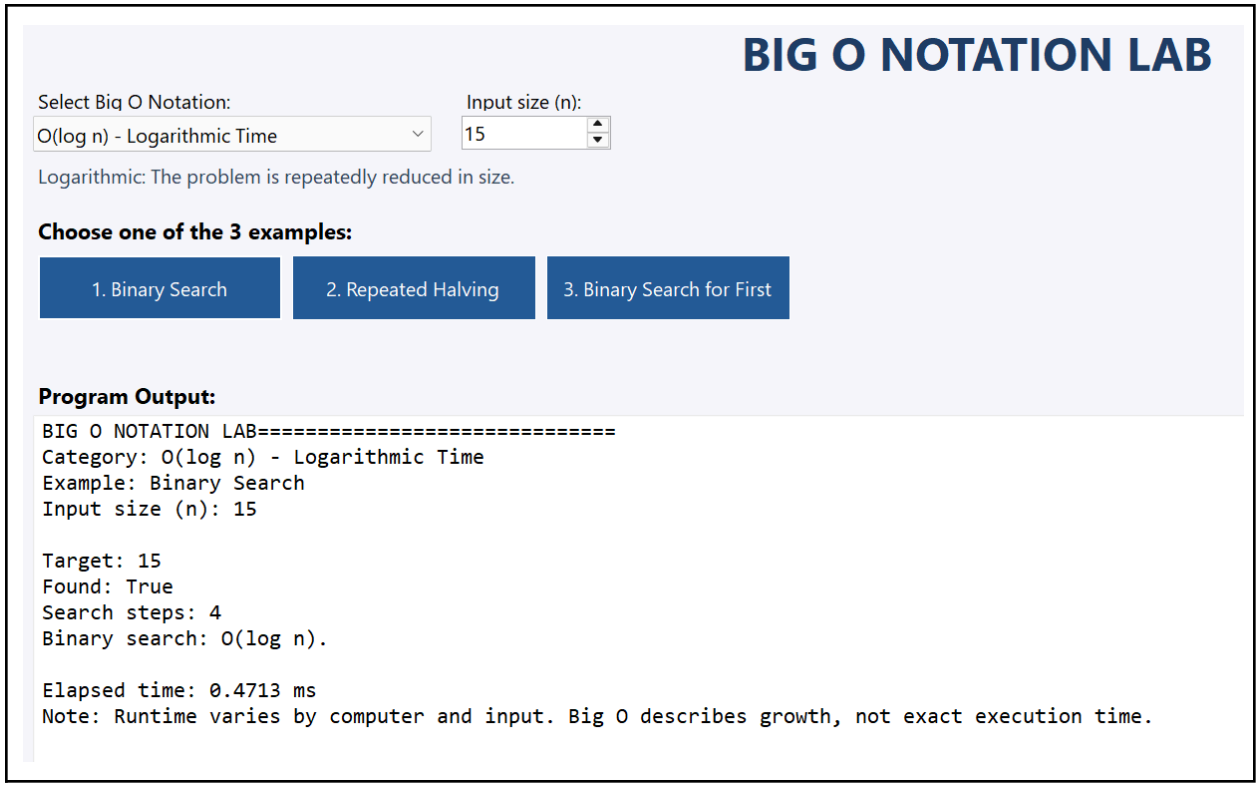

BIG O NOTATION LAB
Select Big O Notation:
O(1) - Constant Time
Input size (n):
40
Constant: The number of steps stays approximately the same.
Choose one of the 3 examples:
1. Access Array Element
2. Get First Element
3. Arithmetic Operation
Program Output:
BIG O NOTATION LAB==============================
Category: O(1) - Constant Time
Example: Get First Element
Input size (n): 40
First element: 1
Only the first element was accessed: O(1).
Elapsed time: 0.0056 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

Example 3


BIG O NOTATION LAB
Select Big O Notation:
O(1) - Constant Time
Input size (n):
35
Constant: The number of steps stays approximately the same.
Choose one of the 3 examples:
1. Access Array Element
2. Get First Element
3. Arithmetic Operation
Program Output:
BIG O NOTATION LAB==============================
Category: O(1) - Constant Time
Example: Arithmetic Operation
Input size (n): 35
10 + 20 = 30
One fixed arithmetic operation: O(1).
Elapsed time: 0.0014 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

O(log n) – Logarithmic Time

Example 1



Example 2


BIG O NOTATION LAB
Select Big O Notation:
O(log n) - Logarithmic Time
Input size (n):
20
Logarithmic: The problem is repeatedly reduced in size.
Choose one of the 3 examples:
1. Binary Search
2. Repeated Halving
3. Binary Search for First
Program Output:
BIG O NOTATION LAB==============================
Category: O(log n) - Logarithmic Time
Example: Repeated Halving
Input size (n): 20
Starting value: 20
Halving steps: 4
Repeated halving: O(log n).
Elapsed time: 0.3784 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

Example 3

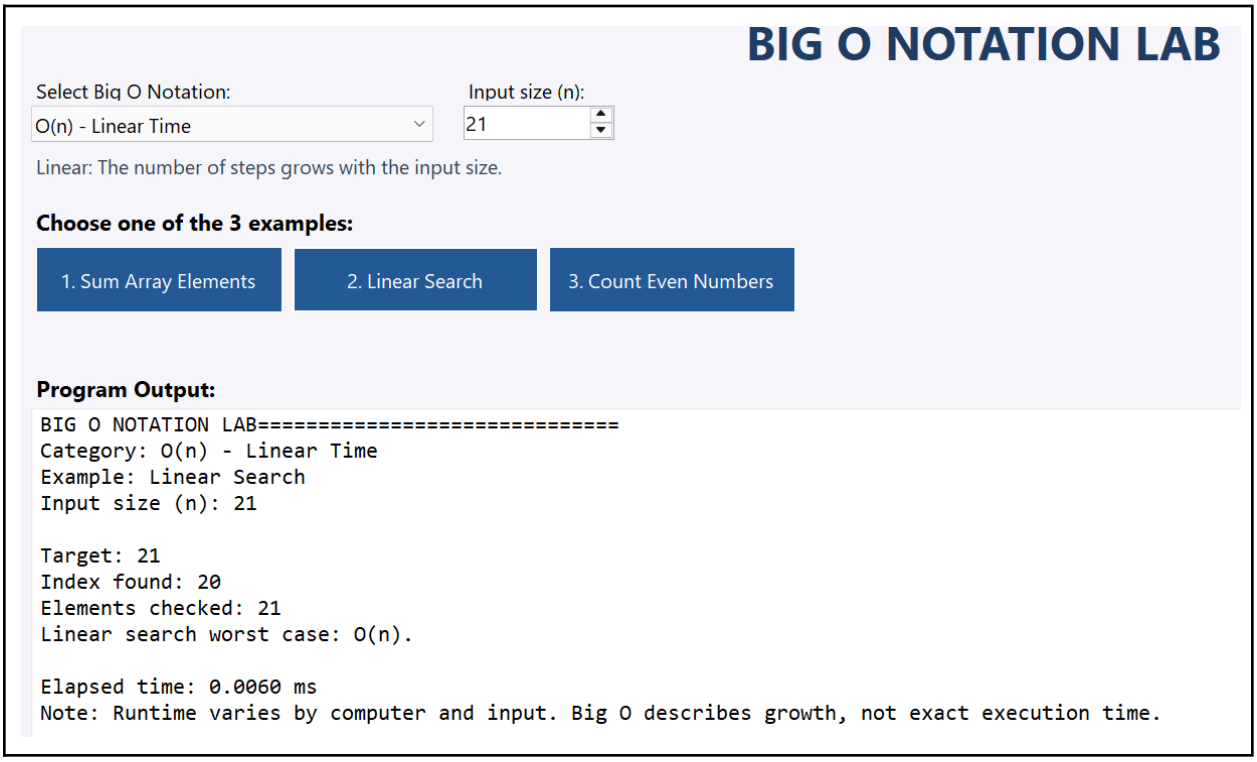


O(n) – Linear Time

Example 1


BIG O NOTATION LAB
Select Big O Notation:
O(n) - Linear Time
Input size (n):
15
Linear: The number of steps grows with the input size.
Choose one of the 3 examples:
1. Sum Array Elements
2. Linear Search
3. Count Even Numbers
Program Output:
BIG O NOTATION LAB==============================
Category: O(n) - Linear Time
Example: Sum Array Elements
Input size (n): 15
Array elements: 15
Sum: 120
Every element is visited once: O(n).
Elapsed time: 0.0062 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

Example 2


BIG O NOTATION LAB
Select Big O Notation:
O(n) - Linear Time
Input size (n):
21
Linear: The number of steps grows with the input size.
Choose one of the 3 examples:
1. Sum Array Elements
2. Linear Search
3. Count Even Numbers
Program Output:
BIG O NOTATION LAB==============================
Category: O(n) - Linear Time
Example: Linear Search
Input size (n): 21
Target: 21
Index found: 20
Elements checked: 21
Linear search worst case: O(n).
Elapsed time: 0.0060 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

Example 3

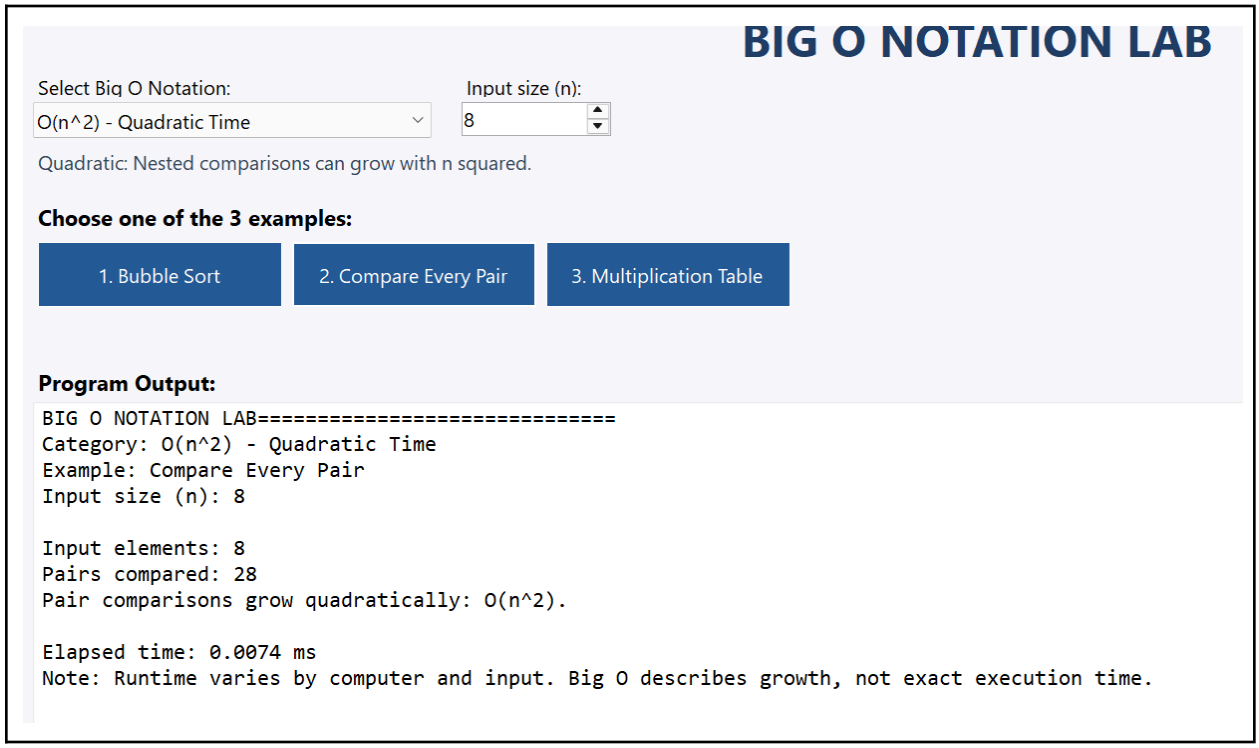

BIG O NOTATION LAB
Select Big O Notation:
O(n) - Linear Time
Input size (n):
30
Linear: The number of steps grows with the input size.
Choose one of the 3 examples:
1. Sum Array Elements
2. Linear Search
3. Count Even Numbers
Program Output:
BIG O NOTATION LAB==============================
Category: O(n) - Linear Time
Example: Count Even Numbers
Input size (n): 30
Numbers checked: 30
Even numbers: 15
One pass through the input: O(n).
Elapsed time: 0.0054 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

O(n²) – Quadratic Time

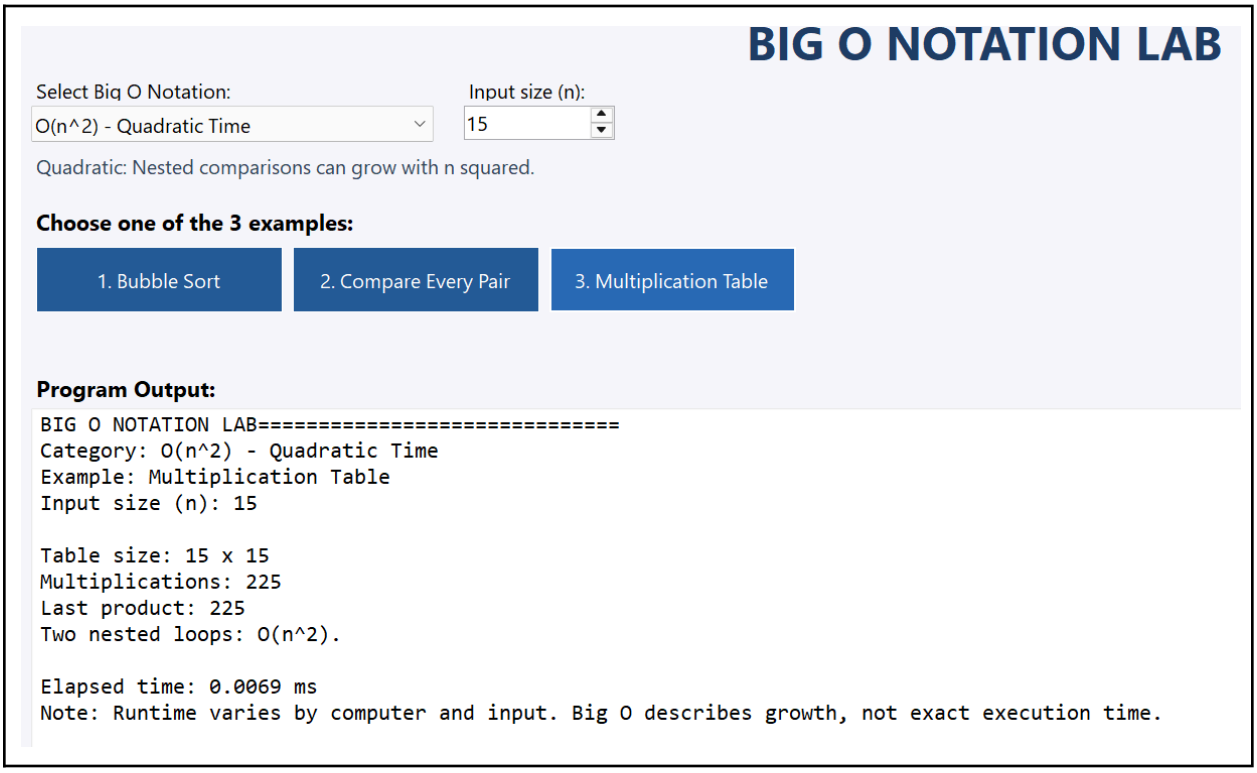
Example 1



Example 2

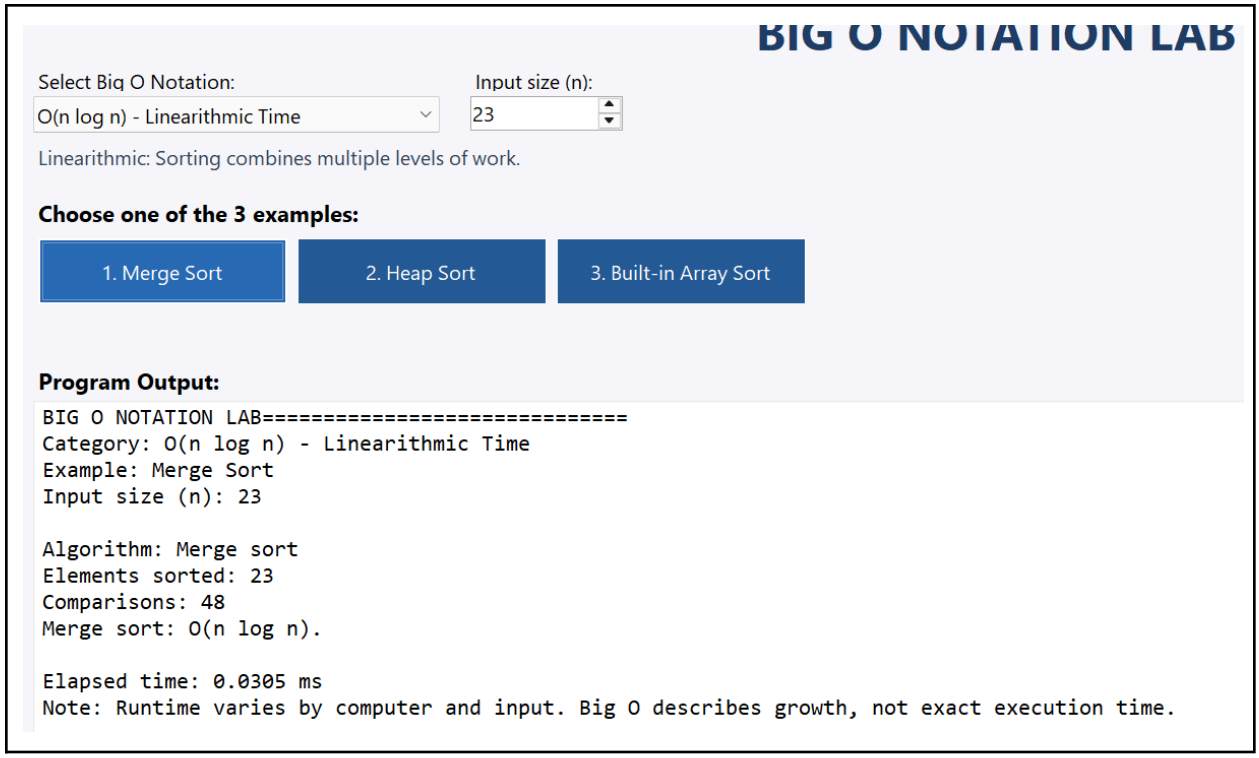

BIG O NOTATION LAB
Select Big O Notation:
O(n^2) - Quadratic Time
Input size (n):
8
Quadratic: Nested comparisons can grow with n squared.
Choose one of the 3 examples:
1. Bubble Sort
2. Compare Every Pair
3. Multiplication Table
Program Output:
BIG O NOTATION LAB==============================
Category: O(n^2) - Quadratic Time
Example: Compare Every Pair
Input size (n): 8
Input elements: 8
Pairs compared: 28
Pair comparisons grow quadratically: O(n^2).
Elapsed time: 0.0074 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

Example 3

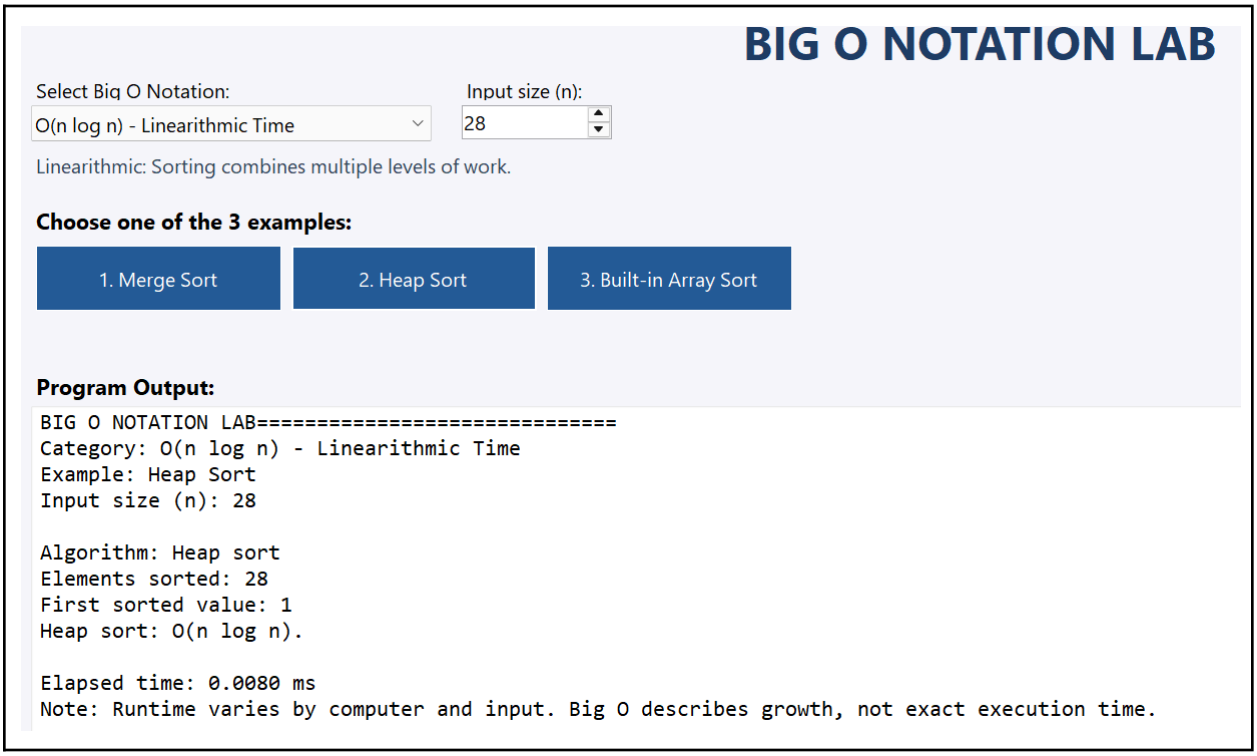

BIG O NOTATION LAB
Select Big O Notation:
O(n^2) - Quadratic Time
Input size (n):
15
Quadratic: Nested comparisons can grow with n squared.
Choose one of the 3 examples:
1. Bubble Sort
2. Compare Every Pair
3. Multiplication Table
Program Output:
BIG O NOTATION LAB==============================
Category: O(n^2) - Quadratic Time
Example: Multiplication Table
Input size (n): 15
Table size: 15 x 15
Multiplications: 225
Last product: 225
Two nested loops: O(n^2).
Elapsed time: 0.0069 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

O(n log n) - Linearithmic Time

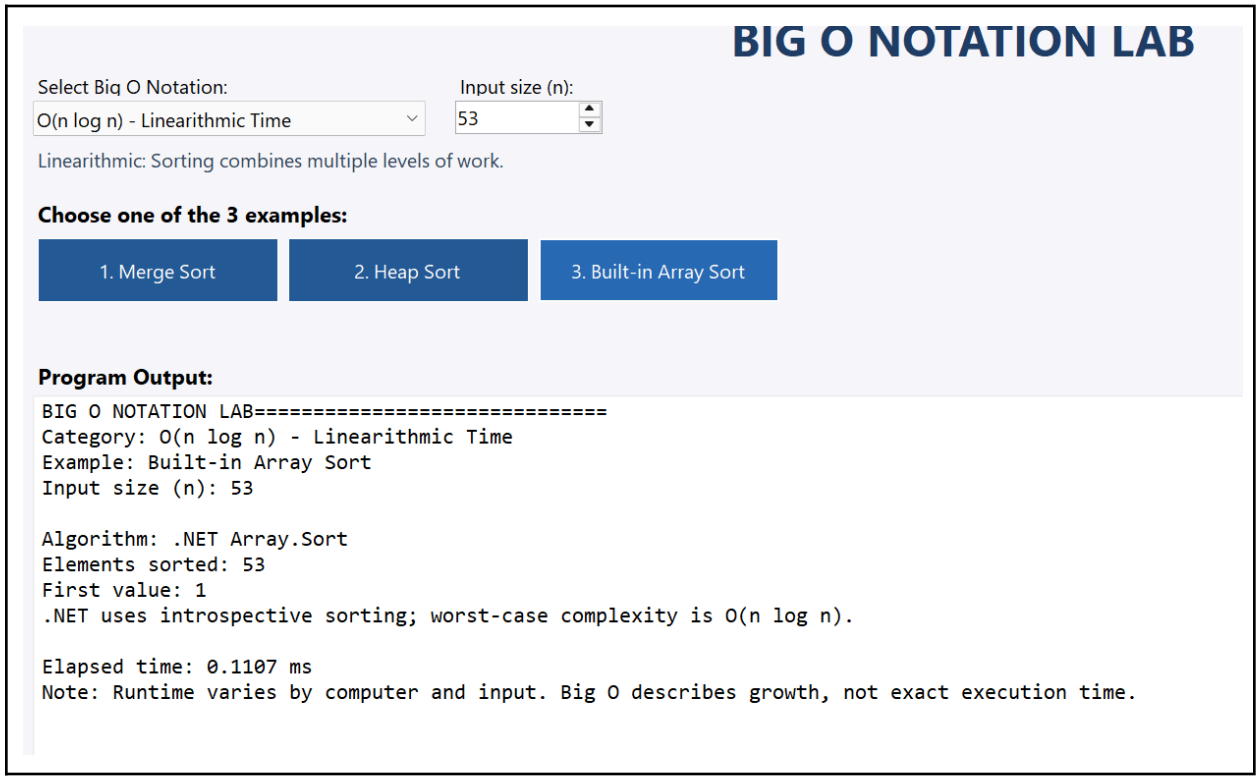
Example 1



Example 2


BIG O NOTATION LAB
Select Big O Notation:
O(n log n) - Linearithmic Time
Input size (n):
28
Linearithmic: Sorting combines multiple levels of work.
Choose one of the 3 examples:
1. Merge Sort
2. Heap Sort
3. Built-in Array Sort
Program Output:
BIG O NOTATION LAB==============================
Category: O(n log n) - Linearithmic Time
Example: Heap Sort
Input size (n): 28
Algorithm: Heap sort
Elements sorted: 28
First sorted value: 1
Heap sort: O(n log n).
Elapsed time: 0.0080 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.

Example 3


BIG O NOTATION LAB
Select Big O Notation:
O(n log n) - Linearithmic Time
Input size (n):
53
Linearithmic: Sorting combines multiple levels of work.
Choose one of the 3 examples:
1. Merge Sort
2. Heap Sort
3. Built-in Array Sort
Program Output:
BIG O NOTATION LAB==============================
Category: O(n log n) - Linearithmic Time
Example: Built-in Array Sort
Input size (n): 53
Algorithm: .NET Array.Sort
Elements sorted: 53
First value: 1
.NET uses introspective sorting; worst-case complexity is O(n log n).
Elapsed time: 0.1107 ms
Note: Runtime varies by computer and input. Big O describes growth, not exact execution time.